



FACULTY OF COMPUTING

SEMESTER 2 2025/2026

SECP3843 - SPECIAL TOPIC IN DATA ENGINEERING

SECTION 2

LECTURER: DR. ARYATI BINTI BAKRI

ASSIGNMENT 1 – TECHNICAL REPORT

TITLE:

TOPIC 13 - DATA ENGINEERING ECOSYSTEM

GROU

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EXECUTIVE SUMMARY

The digital environment of the modern world demands organizations to handle large amounts of information than ever before and thus requires the creation of strong data engineering ecosystems. This technical report discusses the structure, key building blocks and strategic value of the modern data engineering ecosystem. The article identifies the paradigm shift in the monolithic database on-premise to the decoupled, cloud-native architecture that enables scalability and agility. Data ingestion, storage, processing, and orchestration are examined as key components and explored as interdependent functions. The report provides an analysis of the benefits of ecosystem modularity and the complexity of data governance and integration with a comparative analysis, and a case study of a cloud-based academic and talent management platform. The conclusion of the analysis is that sustainable growth requires the use of managed cloud services and decentralized principles of data management. Strategic suggestions are also given, which will include the importance of a strong metadata management and scalable cloud implementation to handle the more complex artificial intelligence and analytics workloads in the enterprise systems.

INTRODUCTION

The fast digitization of the enterprise systems has also redefined data not as a fixed record keeping tool but as a dynamic strategic decision making tool. As a result, the processes of data collection, storage, and transformations of such data have changed tremendously. The data engineering ecosystem is the entire system of tools, platforms, and practices that are created to handle the entire data lifecycle. Its applicability to enterprise systems is hard to exaggerate because a properly designed ecosystem is the underlying infrastructure needed in business intelligence, predictive analytics, and artificial intelligence applications.

Previously, organizations were utilizing separated databases and fixed Extract, Transform, Load (ETL) procedures incapable of handling speedy or unstructured information. The current direction is towards the Modern Data Stack (MDS) which is hugely modular, a cloud-native system that treasures flexibility and scalability. This technical report targets to deconstruct the building blocks of the data engineering ecosystem, the fit of the field in the enterprise architecture, and an evaluation of the current state of the field and its future directions. In this paper, it will attempt to provide a concise overview of how a contemporary data ecosystem will result in operational efficacy and technological progress by concentrating on the structural communication between ingestion, storage, and orchestration.

BACKGROUND

Data engineering ecosystems have also emerged as a result of the necessity to overcome the limitations of the traditional, monolithic data warehousing. According to Kleppmann (2017), the early 2000s were marked by the relational databases that were highly efficient in the transactional processing but failed miserably when they were loaded with much analytics. This shortcoming spurred the creation of distributed computing infrastructures, the most notable being Apache Hadoop, which brought about the data lake notion where organizations can store large volumes of raw data on commodity computers (Zaharia et al., 2016).

The Hadoop ecosystem was however infamously difficult to maintain and did not have the transaction trustworthiness that most enterprises wanted. A recent trend in literature touches on a dramatic architectural change towards solutions based on clouds and the so-called Data Lakehouse paradigm. Armbrust et al. (2021) suggest that the data lakehouse is a convergence between the structural integrity of data warehouses and the flexibility and low cost of data lakes, which has been applied to eliminate data silos. In addition, there has been a clear shift in ETL to ELT (Extract, Load, Transform) approaches in the industry. With the decoupling and extreme elasticity of storage and computing power in clouds, researchers found that loading raw data directly at the destination target and transforming it there greatly decreases latency and pipeline resilience (Inmon and Linstedt, 2014). More recent trends also lay stress on the importance of data orchestration and governance, since the increasing number of divergent tools in the ecosystem require strict metadata management to maintain data quality and provenance in intricate pipelines (Reis & Housley, 2022).

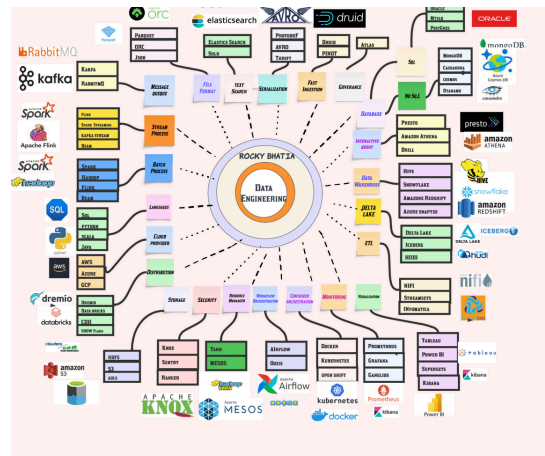


Figure 1 : The Modern Data Engineering Technology Landscape

ANALYSIS AND DISCUSSION

A strong data engineering ecosystem operates based on the flawless coordination of different but highly cooperative technological layers. The architecture is usually initiated by data ingestion where it is constantly removed out of operational databases, external application programming interfaces (APIs), or user interaction events. The current ingestion systems take advantage of both batch processing of historical records and stream processing of real-time analytics. After ingestion, the information goes to the storage layer. Modern architectures strongly prefer cloud-based storage which decouples storage and compute so that an organization can scale resources dynamically according to demand. The raw data is then subjected to intense transformation and modeling once stored. This processing layer purifies, standardizes, and consolidates the data into clean structures that can be further used by analysts and machine learning models. The orchestration tools oversee the entire cycle, schedule tasks, handle dependencies, and recoveries of errors to ensure the pipelines run in predictable ways.

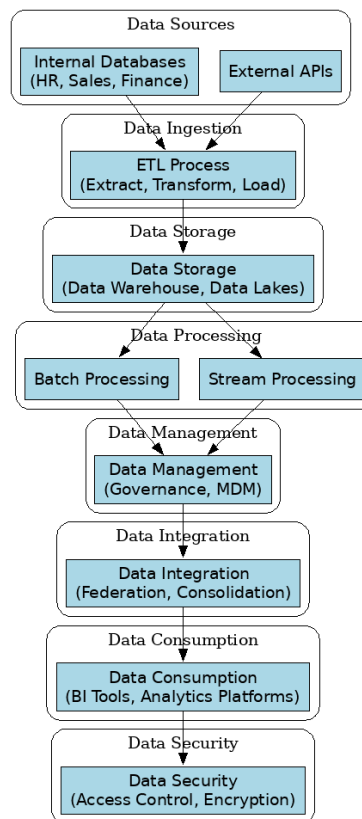


Figure 2 : End-to-End Data Lifecycle in a Data Engineering Ecosystem

The architecture of a scalable platform of student talent verification and academic ecosystem can be regarded as an example to further clarify these concepts. The results of such a system are varied data streams, such as structured user registration information and unstructured activity records and skill endorsement records. An implementation of this platform may be on cloud computing, such as Azure Data Services, to accommodate both low and high computing needs. The data of high-velocity user interaction could be sent to event brokers and academic records is batch-fed at low traffic periods. The data is centrally stored in a data lake which is scalable and is then structured into a definite data warehouse. Transformational logic can then be programmatically used to produce clean datasets that are consumed by automated recommendation models or administrative dashboards. This scale-out design allows the platform to handle traffic spikes when admitting students to the university without any impact on the performance.

The main benefit of this type of ecosystem is the fact that it is highly scalable and flexible. An organization is able to replace individual tools, such as replacing a processing engine, without necessarily completely redesigning the architecture. Moreover, cloud-native operated services have a tremendous impact on relieving engineering teams of the operational workload. There are however significant drawbacks associated with this modularity, in the first place, complexity of integration, and secondly, the danger of vendor lock-in. A dedicated skill set is needed to oversee security measures, access control and data provenance across a dozen software-as-a-service (SaaS) environments.

Strict data governance and quality is the greatest challenge in the contemporary data ecosystems. The more data is democratized within an organization, the more challenging it is to make sure that the results of analytical work are accurate and do not violate privacy policies. In the future, the industry is moving towards Data Mesh architecture that promotes decentralized, domain-based ownership of data, as opposed to a centralized bottleneck of data held by a central data team. Furthermore, embedded artificial intelligence as a part of the ecosystem is becoming a trend, which promises to generate pipelines automatically, transform data into itself, and sense anomalies.

RECOMMENDATIONS

To create a robust and scalable data engineering ecosystem, companies need to focus on shifting to a cloud infrastructure so that the data ecosystem can be elastic and highly scalable. This is the opportunity that enables businesses to decouple storage and compute resources in order to provide allocation of resources more efficiently to address the changing data processing needs. Modern data stack requires agility to save engineers from the heavy maintenance works that are associated with on-premise monolithic systems.

Moreover, the adoption of a unified logical metadata layer is also suggested to manage the growing complexity of different data tools. With the growing ecosystem that incorporates different ingestion and storage systems, a centralized metadata management system is required to ensure that there is a clear understanding of data assets and their lineages. This design eases the discovery of data and gives control to the technical teams to track data movement in the whole lifecycle of end-to-end data.

Lastly, a strict data governance and quality framework is essential to those organizations that are interested in democratizing the access to data. Good governance helps to avoid data silos and maintain consistency and reliability of the information utilized by high-level decision-makers based on the various areas of the business.

CONCLUSION

The current data engineering ecosystem has transitioned from inflexible, monolithic frameworks into a highly modular, cloud-oriented architecture. Companies no longer use traditional ETL, but ELT in order to exploit decoupled storage and compute resources architectures. This segregation is important as it enables the system to dynamically scale and deliver the required power to support the high-velocity data loads without the bottlenecks experienced in the older systems. A key takeaway is the emergence of the Data Lakehouse, which effectively combines the structural robustness of warehouses with the adaptability of data lakes to eliminate information silos. With so many various components involved, it is easy to lose track. To ensure the reliability of our data throughout the process, we must be proactive in the way we handle metadata and oversee the entire process.

The industry is shifting towards a decentralized Data Mesh architecture where prioritizing domain based ownership to relieve centralized bottlenecks in the future. Additionally, the pipeline lifecycle will be improved with artificial intelligence that will autonomously convert data and detect anomalies. The future research needs to be directed towards standardization of such automated tools in order to make various platforms effective in collaborating. The improved interoperability will assist in maintaining the data engineering environments flexible and remain a robust basis of sophisticated analytics and business intelligence.

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APPENDIX

SECP3843 – Special Topic in Data Engineering

Research Logbook – Technical Report

Course: SECP3843 – Special Topic in Data Engineering
Topic: Modern Data Engineering Architecture
Members: Choh Jing Yi (A23CS0296) & Tan Zhi Ming (A23CS0189)

Logbook / Process Record

Date	Activity / Progress	Keywords / Prompts Used	References / Sources	Verification / Notes
2026-03-23	Task Distribution: Met with teammate to finalize the topic. Divided responsibilities: Zhiming (Intro/Background/Analysis/Exec Summary) and Jing Yi (Recommendations/Conclusion).	"Modern Data Engineering Architecture", "Task allocation for group project"	Assignment Instructions	Ensured both members understood the 10% weightage and the April 13 deadline.
2026-03-28	Initial Research & Introduction: Defined the data engineering ecosystem and its strategic importance for AI and BI applications.	"Explain the data engineering ecosystem's role in enterprise systems. Focus on shift to Modern Data Stack."	Reis & Housley (2022)	Aligned the definition with current industry standards regarding the "Modern Data Stack".
2026-03-30	Drafting Background: Researched the evolution from Hadoop and relational databases to the Data Lakehouse paradigm.	"Evolution of data engineering from 2000s relational databases to Data Lakehouses. Mention ELT."	Kleppmann (2017), Armbrust et al. (2021)	Confirmed that the "Data Lakehouse" correctly describes the unification of warehouses and lakes.
2026-04-02	Technical Analysis: Documented the data lifecycle layers, focusing on cloud-native storage and ingestion.	"Layers of modern data architecture: ingestion, storage, processing, and orchestration."	Zaharia et al. (2016), Inmon & Linstedt (2014)	Integrated Figure 2 into the discussion to illustrate the end-to-end data lifecycle.
2026-04-04	Executive Summary: Synthesized the entire report's findings into a concise summary focusing on modularity and cloud-native shifts.	"Executive summary for Modern Data Engineering. Highlight modularity and strategic value."	Research Report Draft	Focused on the "paradigm shift" mentioned in the Analysis section to ensure consistency.
2026-04-05	Drafting Recommendations: Developed strategic suggestions focusing on cloud scalability, metadata management, and proactive governance.	"write in academic writing format... choose 3 recommendation from this article... (Stodder, 2017)"	Stodder, D. (2017) / TDWI	Cross-checked against the "Analysis" section to ensure logical flow. Verified author was Stodder, not Loshin.
2026-04-05	Finalizing Conclusion: Synthesized findings regarding the shift to cloud-native ELT and future trends like Data Mesh.	"write conclusion... Summarize key takeaways and insights. End with forward-looking statement."	Research Report Draft	Used rubric to ensure a forward-looking statement was included regarding automated frameworks.
2026-04-05	Final Review & Appendix: Compiled the AI usage log and formatted the reference list according to APA 7th edition.	"APA 7th citation format", "AI documentation UTM template"	Kleppmann (2017), Reis & Housley (2022)	All sections integrated and citations verified for technical accuracy.

AI Usage & Ethical Statement

In compliance with UTM academic integrity standards, Gemini (AI) was utilized as a research assistant for:

- **Drafting & Refinement:** Polishing the academic tone of the "Recommendations" and "Conclusion" sections.
- **Summarization:** Condensing complex technological concepts into a concise 150-200 word summary for the conclusion.

Verification Steps:

1. **Fact-Checking:** Every recommendation was manually verified against the source article by David Stodder (2017).
2. **Originality Check:** All AI-generated text was rephrased to ensure it represents our own analytical synthesis and passes Turnitin requirements.
3. **Accuracy:** Checked technical definitions (e.g., Data Mesh, ELT) against foundational texts by Reis & Housley (2022).

Tools Consulted

- **GitHub:** Version control and logbook hosting.
- **Gemini:** AI-assisted drafting and summarization.
- **Mendeley/APA Guide:** Citation and reference management.